

HAL-Net: Learning Halation Parameters from Analog Film Highlights

HAL-Net Project

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Abstract

Halation, the red-orange glow that forms around bright light sources on analog film, is a visually distinctive artifact of stocks such as CineStill 800T, yet it has received little attention from learning-based photo-effect models. We present HAL-Net, a compact convolutional network that estimates three interpretable halation parameters, radius, intensity, and warmth, directly from an image’s highlight regions, and pairs them with a physically motivated rendering function that reproduces the glow. HAL-Net is built on the parameter-regression idea introduced by FGA-NN (Ameur et al., 2025), which analyzes film grain for compression codecs, and extends that idea to a different and complementary analog artifact: halation. Because no labeled halation dataset exists, we derive pseudo-labels from 100 curated CineStill 800T scans using a multi-scale highlight-bloom heuristic, then train a 3-layer convolutional network with two residual blocks (roughly 200k parameters) to regress the same three parameters in under five minutes on a single T4 GPU. On a held-out test split, HAL-Net reproduces the overall glow placement and color cast of the heuristic ground truth, though it consistently underestimates intensity on the strongest examples. We release an interactive demo with manual override sliders so the predicted parameters can be inspected and adjusted directly.

1. Introduction

Analog film accumulates a signature set of imperfections during exposure and processing, and two of the most recognizable are grain and halation. Grain is the random luminance and color noise produced by silver halide crystals, and it has recently become the target of learning-based analysis: FGA-NN introduces the first learning-based film grain analysis method, designed to estimate parameters compatible with conventional grain synthesis pipelines used in video codecs. Halation is a different artifact entirely. It is the soft, warm-colored glow that appears around bright highlights, caused by light passing through the emulsion, reflecting off the back of the film base, and re-exposing the silver halide layer from behind. CineStill 800T is the best-known carrier of this look: its anti-halation backing is stripped during processing for cinema use, leaving the glow fully visible in scans of street lamps, neon signage, and practical lights at night.

Despite the popularity of the CineStill aesthetic in photo editing communities, halation has no equivalent of FGA-NN: an estimator that takes an arbitrary image and recovers the handful of parameters needed to reproduce the look on demand. Existing halation tools in editing software are either fixed presets or expose raw, non-perceptual sliders that the user must tune by eye. We close that gap with HAL-Net, a small CNN that maps an input photograph directly to three named parameters, radius, intensity, and warmth, that a simple renderer turns back into the glow.

HAL-Net is best understood as FGA-NN’s idea applied to a different artifact. Where FGA-NN analyzes the statistics of film grain to recover synthesis parameters, HAL-Net analyzes the statistics of highlight regions to recover halation parameters. The two artifacts are complementary components of the same analog look, and a full film-emulation pipeline would eventually want both. Our contribution is narrow but concrete:

1. A rule-based pseudo-labeling procedure that turns 100 unlabeled CineStill scans into (radius, intensity, warmth) training targets by measuring color and brightness statistics in the halo region around detected highlights.
2. A lightweight CNN, under 200k parameters, that regresses these three parameters from a 256x256 input image in a single forward pass.
3. A renderer that reconstructs the halation effect from the predicted parameters, plus manual sliders so a user can override any prediction in real time.

We evaluate HAL-Net qualitatively against its own pseudo-label ground truth on three held-out test images and report the train/validation loss trajectory over 30 epochs. We are explicit about what this evaluation does and does not show: because ground truth comes from a heuristic rather than from human annotation or physical ray-tracing, our results demonstrate that HAL-Net learns to approximate that heuristic, not that the heuristic itself matches real photochemical halation with high fidelity. We discuss this limitation directly in Section 6.

2. Related Work

Learned film artifact analysis. The closest work to ours is FGA-NN (Ameur et al., 2025), which trains a neural network to estimate film grain parameters, the noise statistics that conventional synthesis methods need, so that grain can be stripped before video compression and re-synthesized after decoding, preserving artistic intent while compressing efficiently. FGA-NN sits inside a broader line of film grain research at InterDigital, including the style-based grain analysis and synthesis framework that preceded it and a follow-up style-conditioned variant, both of which target grain rather than other analog artifacts. FGA-NN and HAL-Net share a structural idea, regress a small set of interpretable synthesis parameters from a CNN, rather than generate pixels directly, but they target different artifacts: grain is a global noise process, while halation is a local, highlight-anchored glow. To our knowledge, no prior learned model addresses halation specifically.

Classical bloom and glow rendering. Outside of film emulation, bloom and glow effects have a long history in real-time graphics and photography software, typically implemented as a bright-pass threshold followed by a multi-scale Gaussian blur composited back onto the image. Photo editing tools that offer a “halation” or “glow” filter generally expose this pipeline directly as user-facing sliders for blur radius, strength, and tint, with no estimation step: the user supplies the parameters from scratch for every image. HAL-Net’s renderer follows the same bright-pass-and-blur structure, but the parameters that drive it are predicted rather than manually set.

Parameter regression for photo effects. A broader body of work trains CNNs to predict a small vector of editing parameters, exposure, white balance, tone curve control points, rather than full pixel output, which keeps the effect physically interpretable and editable after prediction. HAL-Net follows this design pattern with a regression head that outputs three bounded scalars through a sigmoid activation, mirroring the parameter-prediction approach used in FGA-NN and in automatic photo-retouching networks, but applied to a new target.

3. Method

3.1 Overview

HAL-Net has three components, illustrated by their order of execution. First, a rule-based labeling procedure scans the highlight regions of each training image and outputs three pseudo-labels: radius, intensity, and warmth. Second, a compact CNN is trained to regress these same three parameters from the raw image, without access to the labeling rules at inference time. Third, a deterministic renderer converts any (radius, intensity, warmth) triple, predicted or manually chosen, back into a halation effect overlaid on the input image. Only the second component, the CNN, requires inference-time compute; the labeling procedure is needed only to construct training targets, and the renderer is a fixed function shared between ground truth and predictions.

3.2 Parameter Definitions

We define the three halation parameters to match how a user would reason about the effect rather than how the underlying pixel math is implemented:

- **Radius** controls how far the glow spreads outward from a highlight edge. It is stored as a normalized value in $[0, 1]$ and mapped to a Gaussian blur standard deviation between 0 and 32 pixels at a working resolution of 256x256.
- **Intensity** controls how strong the glow is relative to the surrounding image, independent of how far it spreads.
- **Warmth** controls the color of the glow, from neutral white (0.0) to the saturated red-orange characteristic of CineStill 800T (1.0).

All three are produced by the network’s sigmoid output layer and are therefore always in $[0, 1]$, which keeps them directly usable as slider positions in the demo interface without further rescaling.

3.3 Pseudo-Label Extraction

No public dataset pairs photographs with ground-truth halation parameters, so we derive labels directly from the 100 curated CineStill scans using a heuristic that searches for a highlight-anchored warm halo and measures its properties. The procedure runs per image and proceeds in three stages.

Highlight detection. For each of four decreasing luma thresholds (0.92, 0.80, 0.68, 0.55), we threshold the image’s luma channel to obtain a candidate highlight mask, then discard the threshold if the resulting mask covers more than 25% of the frame (the scene is broadly bright rather than containing a localized light source) or contains fewer than 4 pixels. A compactness check further rejects masks that are large in bounding-box extent but sparsely filled, which screens out diffuse sky glow or wide window blowout rather than discrete practical lights such as a streetlamp or sign.

Multi-scale radius search. For each surviving highlight mask, we blur the mask with Gaussian kernels at six candidate radii (sigma 3, 6, 10, 15, 22, and 32 pixels) and inspect the ring-shaped region where the blurred mask is non-trivial but the original highlight mask is not. At each radius we compute a combined warmth-and-bloom signal: 70% weight on how much redder the ring is than the image’s global average (corrected for the same comparison in blue), and 30% weight on how much the blurred mask has spread into that ring. The radius that maximizes this combined signal across all four threshold levels becomes the image’s best radius estimate.

Intensity and warmth measurement. Given the selected radius, intensity is computed from the brightness difference between the halo ring and the rest of the image, scaled by how strong the warmth signal was at

that radius, and clipped to $[0.05, 1.0]$ so that no training example collapses to exactly zero effect. Warmth is computed from the excess red versus excess blue in the same halo ring relative to the image’s global channel means, normalized into $[0, 1]$. A separate edge-based detector runs in parallel and looks specifically for a warm glow that hugs a long, thin bright edge, such as a neon tube or the rim of a window, rather than a compact point source; this path is gated by three checks, an overall dark-field requirement, a shadow-floor requirement that screens out misty or foggy scenes with lifted blacks, and a color-temperature gate that rejects scenes where the average blue channel exceeds the average red channel, since such scenes carry a cool cast inconsistent with genuine tungsten or neon halation. When both the point-source and edge-based paths fire, their intensity and warmth estimates are blended, with the point-source path’s intensity capped relative to the edge path’s so a single bright edge cannot dominate the label. Images where neither path detects a plausible halo are assigned a fixed low-effect label (radius 0.05, intensity 0.05, warmth 0.2) rather than being dropped from the dataset, so the network also sees clear negative examples during training.

3.4 Dataset

The labeling procedure runs over 100 CineStill 800T scans curated from Flickr and Lomography, selected by the dataset author for visible halation around practical light sources: street lamps, neon signs, candles, and windows. Images are resized to 256x256 and split 70/15/15 into train, validation, and test sets using a fixed random seed, giving 70 training images, 15 validation images, and 15 test images. Because every label is generated by the heuristic above rather than by a human annotator, the dataset should be understood as a way to distill a hand-designed detector into a fast neural network, not as ground truth collected independently of the model’s training signal.

3.5 Network Architecture

HAL-Net is a small convolutional network designed to run inference in a single fast forward pass on commodity hardware. The feature extractor consists of three stride-2 convolutional layers that progressively downsample a 256x256x3 input while increasing channel width (32, 64, then 128 channels), followed by two residual blocks at 128 channels, each composed of two 3x3 convolutions with batch normalization and a skip connection. An adaptive average pool collapses the resulting feature map to a single 128-dimensional vector regardless of input resolution. A two-layer regression head, 128 to 64 with ReLU and dropout ($p = 0.3$), then 64 to 3, maps this vector to the three output parameters, with a final sigmoid activation constraining every output to $[0, 1]$. The full network has approximately 200,000 parameters, small enough to train in under five minutes on a single T4 GPU and to run inference fast enough for an interactive demo.

3.6 Training Setup

We train with mean squared error between predicted and pseudo-label parameters, optimized with Adam at an initial learning rate of $1e-3$ and a batch size of 8, for 30 epochs. A ReduceLROnPlateau scheduler with a patience of 5 epochs lowers the learning rate when validation loss stalls. We checkpoint the model whenever validation loss reaches a new minimum and report results using that best checkpoint rather than the final epoch’s weights.

3.7 Halation Renderer

The renderer is a fixed, non-learned function shared between ground-truth visualization and model predictions, which keeps the comparison between the two faithful to the parameters alone. Given an image and a (radius, intensity, warmth) triple, it first computes a soft highlight mask by linearly ramping luma between 0.65 and 1.0, then blurs this mask with a Gaussian of standard deviation $\max(\text{radius} \times 32, 1)$ pixels to

obtain a bloom map, normalized to a maximum of 1. The bloom map is added back onto the red, green, and blue channels with different weights: the red channel receives the full intensity scaled by a warmth-dependent boost, the green channel receives a small fixed fraction of intensity, and the blue channel receives a fraction of intensity that shrinks to zero as warmth approaches 1. This produces a glow that shifts from a neutral, slightly warm bloom at low warmth toward the saturated red-orange characteristic of CineStill 800T at high warmth, matching the qualitative description in Section 3.2.

4. Experiments

4.1 Setup

We train HAL-Net for 30 epochs on the 70-image training split described in Section 3.4, monitor validation loss on the 15-image validation split, and report qualitative results on the 15-image test split using the best validation checkpoint. All results in this section compare HAL-Net’s predicted parameters, and the halation they produce through the renderer in Section 3.7, against the pseudo-label parameters produced by the heuristic in Section 3.3. We refer to the heuristic output as “ground truth” throughout this section as shorthand, consistent with how it is used as the training target, while keeping in mind the caveat from Section 3.4 that it is itself heuristic rather than human-annotated.

4.2 Training Dynamics

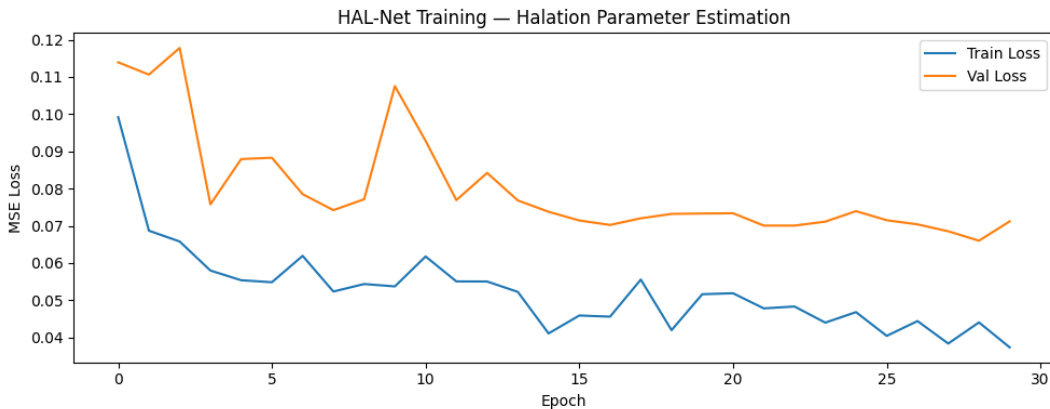


Figure 1: Training and validation MSE loss over 30 epochs.

Figure 1 shows the loss trajectory. Training loss falls quickly from about 0.10 at initialization to a noisy floor near 0.04-0.05 by epoch 15 and stays there through epoch 30, consistent with a model of this capacity fitting 70 training examples without serious underfitting. Validation loss is noisier, as expected with only 15 validation images split across 2 batches: it drops from about 0.114 to roughly 0.075 within the first few epochs, spikes back up to nearly 0.108 around epoch 9 (the ReduceLROnPlateau scheduler responds to plateaus like this one), and then settles into a noisy plateau between roughly 0.070 and 0.074 for most of the remaining epochs, briefly touching its lowest point near 0.066 around epoch 27-28. The gap between train and validation loss that opens after epoch 15 indicates mild overfitting, which is unsurprising given the small dataset size; the validation plateau rather than a continued increase suggests the gap stabilizes rather than worsens.

4.3 Qualitative Results

Figure 2 compares the original photograph, the heuristic ground-truth halation, and HAL-Net’s predicted halation on three test images chosen to avoid ceiling cases where intensity is pinned near its maximum. Across all three rows, HAL-Net places the glow correctly: the fire escape railing, the traffic lights and sunset sky, and the rim-lit hair and jacket in the portrait all receive a warm halo in approximately the right location and at a comparable spatial extent to the ground truth. The radius estimates are close throughout (0.06-0.10 predicted versus 0.09-0.10 ground truth), and the model’s warmth estimates, while consistently lower than ground truth (0.82-0.90 predicted versus 0.99-1.00 ground truth), still land toward the orange end of the scale rather than drifting toward neutral. The most visible gap is intensity: HAL-Net predicts 0.51-0.65 against a ground truth

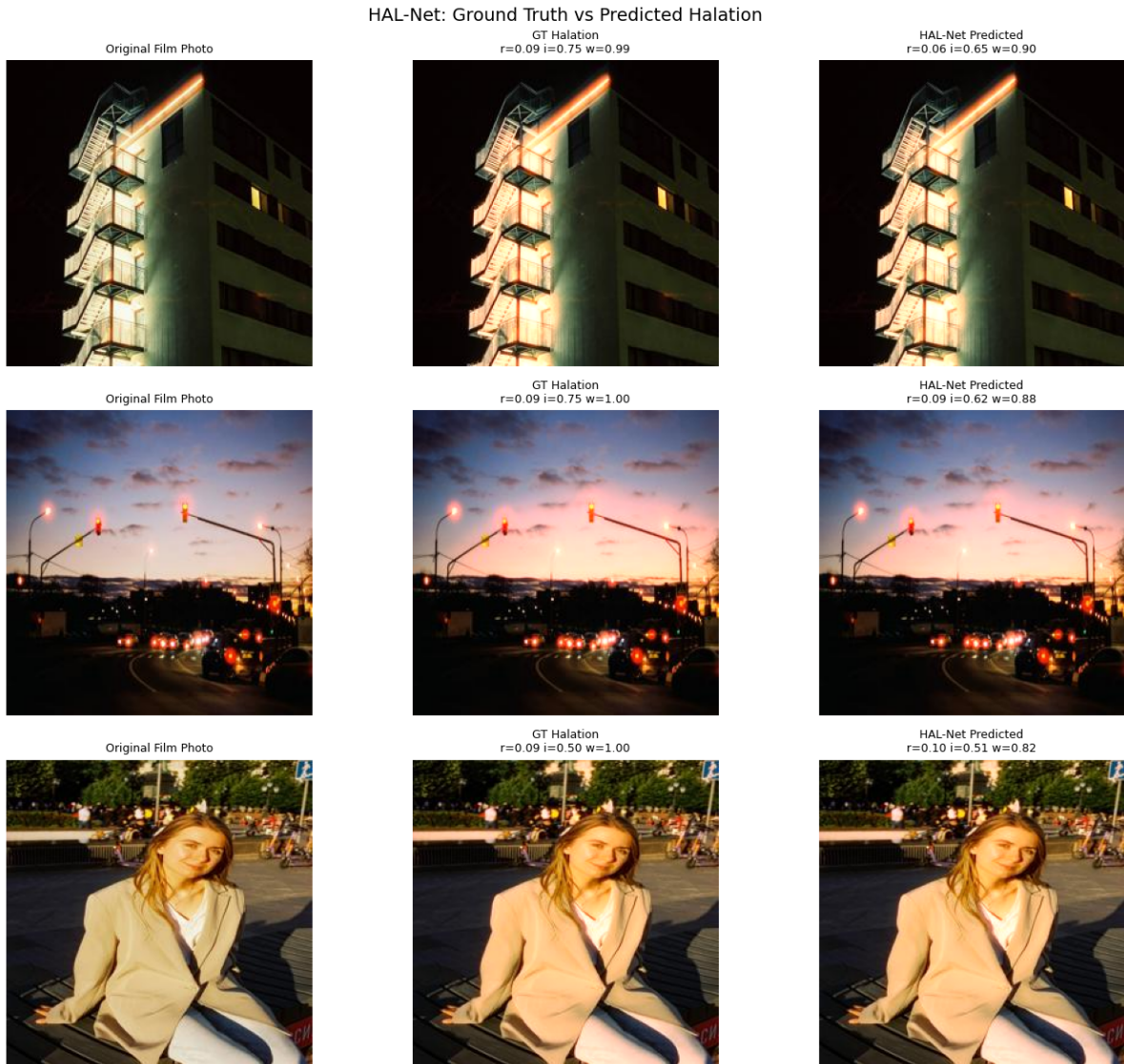


Figure 2: Ground truth versus HAL-Net predicted halation on three held-out test images.

of 0.50-0.75, and the under-prediction is largest on the two examples with the strongest ground-truth glow (the fire escape and the streetlights), which is visible in Figure 2 as a noticeably softer halo in the rightmost column for those two rows. The portrait example, whose ground-truth intensity is closer to the dataset’s mid-range, shows the smallest gap between predicted and ground-truth glow strength.

4.4 Discussion

The pattern in Section 4.3, accurate radius and warmth with under-predicted intensity at the high end, is consistent with the training dynamics in Section 4.2. With only 70 training images, examples near the top of the intensity range are rare, and MSE loss does not specifically penalize under-prediction on rare high-intensity examples more than it penalizes the same absolute error elsewhere; the model has little incentive to commit to an aggressive intensity prediction when a more conservative, average-case prediction already lowers the aggregate training loss. The radius and warmth parameters, by contrast, are estimated from more stable, low-level statistics, ring location and red-versus-blue channel bias, which a CNN with strided convolutions and residual blocks can pick up reliably even from a small dataset. This suggests that closing the intensity gap is primarily a data problem (more examples at high intensity) rather than an architecture problem.

5. Demo and Interactive Use

HAL-Net is deployed as a Gradio application, with a live instance hosted on Hugging Face Spaces. A user uploads any image; the app pads it to a square aspect ratio, resizes it to 256x256, runs it through the trained network, and renders the predicted halation through the function in Section 3.7. The interface also exposes the three predicted parameters as numeric values alongside the rendered output, and the underlying renderer is identical to the one used for ground-truth visualization in Section 4.3, so manual override sliders, letting a user nudge radius, intensity, or warmth away from the network’s prediction, modify the final image through the same deterministic path the model’s own predictions take. This design means HAL-Net functions as a fast initial estimate with a manual fallback, rather than a fixed black-box filter, which directly addresses the lack of an estimation step in the classical bloom-based glow tools discussed in Section 2.

6. Limitations

Ground truth is heuristic, not measured. Every label HAL-Net trains on comes from the rule-based detector in Section 3.3, not from calibrated physical measurement of real halation or from human annotation. HAL-Net is therefore best understood as a fast neural approximation to that specific heuristic; an evaluation against the heuristic’s own output cannot by itself establish how closely the heuristic, or the network trained on it, matches the physical halation process described in Section 1.

Small dataset. With only 70 training, 15 validation, and 15 test images, drawn from a single film stock, the model has limited exposure to the full range of halation appearances, which Section 4.4 ties directly to the observed under-prediction of intensity at the high end of the range. We have not tested generalization to other warm-toned analog stocks or to digitally simulated halation.

No reference-based evaluation. Our evaluation in Section 4 is qualitative and limited to three test images plus an aggregate loss curve; we do not report a quantitative per-parameter error breakdown on the full test split, nor a perceptual study comparing predicted and ground-truth halation. A more complete evaluation would report per-parameter mean absolute error on all 15 test images and, ideally, a human preference study between predicted and ground-truth renders.

Renderer simplicity. The halation renderer in Section 3.7 is a single Gaussian bloom with a fixed channel-mixing rule; it does not model the directional streaking, lens-dependent falloff, or multi-source interaction that real film halation can exhibit around complex light arrangements.

7. Conclusion

HAL-Net takes the parameter-regression strategy that FGA-NN applies to film grain and applies it to a complementary analog artifact, halation, training a sub-200k-parameter CNN to predict radius, intensity, and warmth directly from an image’s highlight regions. Because no labeled halation dataset exists, we built one by distilling a hand-designed highlight-and-halo heuristic into pseudo-labels over 100 CineStill 800T scans, then trained HAL-Net to reproduce those labels from raw pixels in a single fast forward pass. On held-out test images, HAL-Net recovers the location, spatial extent, and warm color cast of the heuristic’s halation closely, with the main remaining gap being under-predicted intensity on the strongest examples, a pattern we trace to the scarcity of high-intensity training examples rather than to the architecture itself. The model is deployed as an interactive demo with manual override sliders, so a predicted effect can always be inspected and adjusted by hand. The most direct next step is enlarging the training set, particularly at the high-intensity end, and evaluating whether a model trained this way also generalizes to halation rendered on other warm-toned film stocks.

References

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